One framework, three different questions

Learning style describes training. Domain describes the data or application. Task describes the desired output.

LEARNING STYLE

Supervised: learn from target labels.
Unsupervised: discover structure without external task labels.

APPLICATION DOMAIN

Tabular: records and columns.
Vision: images. NLP: language.
Generative AI: content creation.

TASK

Predict a value or category.
Group or compress examples.
Generate new content.

COMMON DOMAIN FIT

A filled circle marks a common fit; an open circle means use on extracted numeric features.

Algorithm / model family	Tabular data	Computer vision	NLP	Generative AI
Linear regression	●			
Decision tree	●			
Random forest	●			
Gradient boosting	●			
CNN classifier		●		
K-means	●	○	○	
PCA	●	○	○	
GPT-style transformer			●	●
GAN		●		●
Diffusion model		●		●

Domains overlap: an image generator belongs to both computer vision and generative AI. A CNN is an architecture; supervised describes the classifier training shown here. [1-8]

Learn from labeled examples

Supervised learning maps inputs to known targets. Classification predicts a category; regression predicts a number.

01 Linear regression

Supervised

How: Fits a weighted sum of features to predict a number.

Use: Estimate house prices from area, location, and age.

Domain: Tabular | Interpretable baseline; limited by a linear relationship. [1]

02 Decision tree

Supervised

How: Learns feature questions that split labeled data into purer groups.

Use: Classify fruit using extracted color and shape measurements.

Domain: Tabular | Readable rules; an unrestricted tree can overfit. [1]

03 Random forest

Supervised

How: Trains randomized trees on resampled data, then votes or averages.

Use: Predict customer churn using account and usage records.

Domain: Tabular | Combines trees in parallel to reduce variance. [2]

04 Gradient boosting

Supervised

How: Adds weak models in sequence to reduce the current prediction loss.

Use: Predict insurance claim costs from customer records.

Domain: Tabular | Strong nonlinear predictor; tuning helps control overfitting. [2]

05 CNN classifier

Supervised in this use

How: Learns shared image filters, then maps visual features to labels.

Use: Identify manufacturing defects in labeled product photographs.

Domain: Computer Vision | Learns spatial features; CNNs also appear inside generators. [5]

Relationship: One decision tree gives one rule path. Random forests combine randomized trees in parallel; gradient boosting adds models sequentially. [1-2]

Find structure or generate content

K-means groups examples; PCA compresses features. GPT, GANs, and diffusion models learn patterns for content generation.

06 K-means

Unsupervised

How: Alternates assigning points to the nearest center and updating centers.

Use: Group customers by purchase frequency and spending.

Domain: Tabular; image/text features | Choose K and scale features; clusters need interpretation. [3]

07 PCA

Unsupervised

How: Projects centered data onto directions that retain the most variance.

Use: Compress correlated network traffic measurements for analysis.

Domain: Tabular; image/text features | Reduces dimensions; does not itself predict labels. [4]

08 GPT-style transformer

Unsupervised / self-supervised*

How: Uses attention and prior tokens to predict the next token repeatedly.

Use: Draft support replies or summarize technical documents for review.

Domain: NLP; Generative AI | Creates text; fluent output can still contain errors. [6]

09 GAN

Unsupervised, basic form*

How: A generator makes samples while a discriminator learns real vs. fake.

Use: Create synthetic product images to expand a training collection.

Domain: Computer Vision; Generative AI | Adversarial learning; limited variety can occur during training. [7]

10 Diffusion model

Unsupervised / self-supervised*

How: Learns to remove added noise; generation reverses noise step by step.

Use: Generate concept artwork; Stable Diffusion is a text-conditioned example.

Domain: Computer Vision; Generative AI | Iterative denoising; conditioning can guide the output. [8]

***Training nuance:** Self-supervision creates targets from the data itself. GPT may later receive supervised fine-tuning. Conditional GANs and diffusion models can use labels or paired text-image data; their training style depends on setup. [6-8]

Portfolio context and source notes

Artifact purpose: communicate model selection across data types, learning styles, and tasks. This is a conceptual framework, not a benchmark.

MODEL SELECTION IN PRACTICE

For labeled network records, begin with a tabular predictor. For unlabeled traffic groups, consider clustering. For visual inspection, consider a CNN classifier. For text or image creation, consider a generative model and review outputs for accuracy, bias, privacy, and suitability.

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[4] scikit-learn developers. (n.d.). Decomposing signals in components.
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[5] TensorFlow. (n.d.). Convolutional neural network (CNN).
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[6] Brown, T. B., et al. (2020). Language models are few-shot learners. arXiv.
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[8] Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. arXiv.
https://arxiv.org/abs/2006.11239

Reference numbers link explanations to sources. Use cases are illustrative applications, not claims about a named organization or measured results. The domain matrix is a guide, not an exhaustive list of what each method can do.